

Anaemia

1. Anaemia refers to a state in which the level of haemoglobin in the blood is below the normal range appropriate for age and sex.
2. Other factors, including pregnancy and altitude, also affect haemoglobin levels and must be taken into account when considering whether an individual is anaemic.
3. The clinical features of anaemia reflect diminished oxygen supply to the tissues.
4. A rapid onset of anaemia (e.g. due to blood loss) causes more profound symptoms than a gradually developing anaemia.
5. Individuals with cardio-respiratory diseases are more susceptible to symptoms of anaemia.
6. Around 30 % of the total world population is anaemic and half of these, some 600 million people, have iron deficiency.
7. The classification of anaemia by the size of the red cells (MCV) indicates the likely cause.
8. Red cells in the bone marrow must acquire a minimum level of haemoglobin before being released into the blood stream.
9. Whilst in the marrow compartment, red cell precursors undergo cell division driven by erythropoietin.
10. If red cells cannot acquire haemoglobin at a normal rate, they will undergo more divisions than normal and will have a low MCV when finally released into the blood.
11. The MCV is low because component parts of the haemoglobin molecule are not fully available: that is, iron in iron deficiency, globin chains in thalassaemia, haem ring in congenital sideroblastic anaemia and, occasionally, poor iron utilisation in the anaemia of chronic disease.
12. In megaloblastic anaemia the biochemical consequence of vitamin B₁₂ or folate deficiency

is an inability to synthesise new bases to make DNA.

13. A similar defect of cell division is seen in the presence of cytotoxic drugs or haematological disease in the marrow such as myelodysplasia.
14. In these states, cells haemoglobinise normally but undergo fewer cell divisions, resulting in circulating red cells with a raised MCV.
15. The red cell membrane is composed of a lipid bilayer which will freely exchange with the plasma pool of lipid.
16. Conditions such as liver disease, hypothyroidism, hyperlipidaemia and pregnancy are associated with raised lipids and may also cause a raised MCV.
17. Reticulocytes are larger than mature red cells, so when the reticulocyte count is raised—for example, in haemolysis—this may also increase the MCV.

Clinical assessment

1. Iron deficiency anaemia is the most common type of anaemia world-wide.
2. A thorough gastrointestinal history is important, looking in particular for symptoms of blood loss.
3. Menorrhagia is a common cause of anaemia in pre-menopausal females, so women should always be asked about their periods.
4. A dietary history should assess the intake of iron and folate which may become deficient in comparison to needs (e.g. in pregnancy or during periods of rapid growth).

Past medical history

Diseases known to be associated with anaemia,

- Rheumatoid arthritis (the anaemia of chronic disease)
- Previous surgery (e.g. resection of the stomach)

or small bowel which may lead to malabsorption of iron and/or vitamin B₁₂).

Family history and ethnic background

- It may raise suspicion of haemolytic anaemias such as the haemoglobinopathies and hereditary spherocytosis.
- Pernicious anaemia may also be familial.

A drug history : It may reveal the ingestion of drugs which cause blood loss (e.g. aspirin and antiinflammatory drugs), haemolysis or aplasia.

Investigations

1. Schemes for the investigation of anaemias are often based on the size of the red cells, which is most accurately indicated by the MCV in the FBC. Commonly the presence of anaemia.
2. A normal MCV (normocytic anaemia) suggests either acute blood loss or the anaemia of chronic disease (ACD)
3. A low MCV (microcytic anaemia) suggests iron deficiency or thalassaemia
4. A high MCV (macrocytic anaemia) suggests vitamin B₁₂ or folate deficiency or myelodysplasia.

Classification of Anaemias

Several types of classifications of anaemias have been proposed. Two of the widely accepted classifications are based on the pathophysiology and morphology

Pathophysiologic Classification

Depending upon the pathophysiologic mechanism, anaemias are classified into 3 groups:

I. Anaemia due to blood loss -This is further subdivided into 2 types:

- A. Acute post-haemorrhagic anaemia
- B. Anaemia of chronic blood loss

II. Anaemia due to impaired red cell formation

-A disturbance due to impaired red cell production from various causes may produce anaemia. These are as under :

- A. Cytoplasmic maturation defects
 1. Deficient haem synthesis : iron deficiency anaemia

2. Deficient globin synthesis : thalassaemic syndromes

B. Nuclear maturation defects

Vitamin B₁₂ and/or folic acid deficiency: megaloblastic anaemia

C. Haematopoietic stem cell proliferation and differentiation abnormality e.g.

1. Aplastic anaemia
2. Pure red cell aplasia

D. Bone marrow failure due to systemic diseases (anaemia of chronic disorders) e.g.

1. Anaemia of inflammation/infections, disseminated malignancy
2. Anaemia in renal disease
3. Anaemia due to endocrine and nutritional deficiencies (hypometabolic states)
4. Anaemia in liver disease

E. Bone marrow infiltration e.g.

1. Leukaemias
2. Lymphomas
3. Myelosclerosis
4. Multiple myeloma

F. Congenital anaemia e.g.

1. Sideroblastic anaemia
2. Congenital dyserythropoietic anaemia.

The term hypoproliferative anaemias is also used to denote impaired marrow proliferative activity and includes 2 main groups : hypoproliferation due to iron deficiency and that due to other hypoproliferative disorders; the latter category includes anaemia of chronic inflammation/infection, renal disease, hypometabolic states, and causes of bone marrow failure.

III. Anaemia due to increased red cell destruction (haemolytic anaemias) This is further divided into 2 groups :

A. Intracorpuscular defect (hereditary and acquired haemolytic anaemias).

Morphologic Classification

Based on the red cell size, haemoglobin content and red cell indices, anaemias are classified into 3 types:

1. **Microcytic, hypochromic** - MCV, MCH, MCHC are all reduced e.g. in iron deficiency anaemia and in certain non-iron deficient anaemias (sideroblastic anaemia, thalassaemia, anaemia of chronic disorders).
2. **Normocytic, normochromic** - MCV, MCH, MCHC are all normal e.g. after acute blood loss, haemolytic anaemias, bone marrow failure, anaemia of chronic disorders.
3. **Macrocytic** - MCV is raised e.g. in megaloblastic anaemia due to deficiency of vitamin B₁₂ or folic acid.

High yield information

- * Anaemia is reduced haemoglobin concentration in blood below the lower limit of the normal for the age and sex of the individual.
- * Anaemia causes lowered oxygen-carrying capacity and eventually tissue hypoxia and impaired function.
- * Major symptoms of anaemia are tiredness, weakness and lethargy and main signs are pallor, hyperdynamic circulation, and impaired function of CNS and kidney.
- * A suspected case of anaemia is investigated by haemoglobin estimation, peripheral blood smear examination, and complete blood counts including reticulocyte count and ESR.
- * Bone marrow aspiration and trephine biopsy are done to confirm and type the anaemia.
- * Anaemias are classified based on pathophysiology (into anaemia due to blood loss, impaired red cell destruction) or morphology (into microcytic hypochromic, macrocytic, normocytic normochromic).

Classification of anaemias in brief

- A. Pathophysiologic
 - I. Anaemia due to increased blood loss
 - a) Acute post-haemorrhagic anaemia
 - b) Chronic blood loss
 - II. Anaemias due to impaired red cell production
 - a) Cytoplasmic maturation defects
 1. Deficient haem synthesis :

Iron deficiency anaemia

2. Deficient globin synthesis :

Thalassaemic syndromes

- b) Nuclear maturation defects
 - Vitamin B₁₂ and /or folic acid deficiency :**
 - Megaloblastic anaemia
 - c) Defect in stem cell proliferation and differentiation
 1. Aplastic anaemia
 2. Pure red cell aplasia
 - d) Anaemia of chronic disorders
 - e) Bone marrow infiltration
 - f) Congenital anaemia
- III. Anaemias due to increased red cell destruction (Haemolytic anaemias)
 - A. Extrinsic (extracorporeal) red cell abnormalities
 - B. Intrinsic (intracorporeal) red cell abnormalities
 - C. Morphologic
 - I. Microcytic, hypochromic
 - II. Normocytic, Normochromic
 - III. Macrocytic, normochromic

Iron deficiency anaemia

It occurs when iron losses or physiological requirements exceed absorption.

Aetiology

A. Blood loss :

1. The most common explanation in men and postmenopausal women is gastrointestinal blood loss
2. In developed countries one-third of premenopausal women have low iron stores but only 3% display iron-deficient haematopoiesis.
3. Very rarely, chronic haemoptysis or haematuria may cause iron deficiency.

B. Malabsorption :

1. Gastric acid is required to release iron from food and helps to keep iron in the soluble ferrous state
2. Hypochlorhydria in the elderly or that due to

drugs such as proton pump inhibitors may contribute to the lack of iron availability from the diet, as may previous gastric surgery.

3. Iron is absorbed actively in the upper small intestine and hence can be affected by coeliac disease

C. Physiological demands

1. At times of rapid growth, such as infancy and puberty, iron demands increase and may outstrip absorption.

Investigation of the cause

1. It depends upon the age and sex of the patient, as well as the history and clinical findings.
2. The upper and lower gastrointestinal tract should be investigated by endoscopy or barium studies
3. Serum antiendomysial antibodies and possibly a duodenal biopsy are indicated to detect coeliac disease in iron-deficient patients who have features of malabsorption or recurrent deficiency in the absence of other explanations, or who are young men with normal diet or young women with normal menstruation and diet.
4. In the tropics, stool and urine should be examined for parasites
5. *Confirmation of iron deficiency* plasma ferritin is a measure of iron stores in tissues and is the best single test to confirm iron deficiency.
6. Plasma iron and total iron binding capacity (TIBC) are measures of iron availability
7. Levels of transferrin, the binding protein for iron

Anaemia of chronic disease (ACD)

1. This is a common type of anaemia, particularly in hospital populations.
2. It occurs in the setting of chronic infection, chronic inflammation or neoplasia.
3. The anaemia is not related to bleeding, haemolysis or marrow infiltration, is mild, with Hb in the range of 85-115g/L, and is usually associated with a normal MCV (normocytic, normochromic), though this may be reduced in long-standing inflammation.

4. The serum iron is low but iron stores are normal or increased, as indicated by the ferritin or stainable marrow iron.

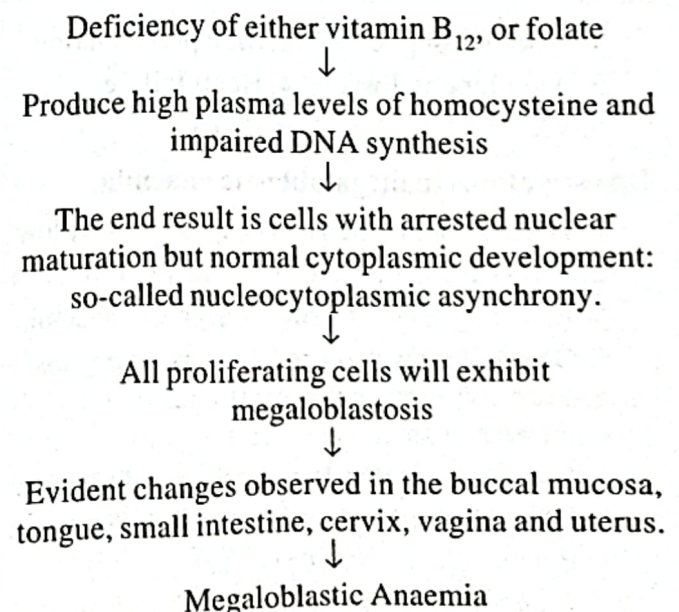
Pathogenesis

1. It has recently become clear that the key regulatory protein that accounts for the findings characteristic of ACD is hepcidin, which is produced by the liver.
2. High levels of production are encouraged by proinflammatory cytokines, especially IL-6. Hepcidin binds to ferroportin on the membrane of iron exporting cells, such as small intestinal enterocytes and macrophages, internalising the ferroportin and thereby inhibiting the export of iron from these cells into the blood (and hence to the main target cells and proteins of iron).
3. The iron remains trapped inside the cells in the form of ferritin, levels of which are therefore normal or high in the face of significant anaemia.
4. Inhibition or blockade of hepcidin is a likely target for potential treatment of this form of anaemia.

Megaloblastic anaemia

Aetiology : This results from a deficiency of vitamin B₁₂, or folic acid or from disturbances in folic acid metabolism.

Pathogenesis



Dietary deficiency

This only occurs in strict vegans but the onset of clinical features can occur at any age between 10 and 80 years.

Gastric factors

Release of vitamin B₁₂ from the food requires normal gastric acid and enzyme secretion, and this is impaired by hypochlorhydria in elderly patients or following gastric surgery.

Pernicious anaemia

This is an autoimmune disorder in which the gastric mucosa is atrophic, with loss of parietal cells causing intrinsic factor deficiency.

Small bowel factors

One-third of patients with pancreatic exocrine insufficiency fail to transfer dietary vitamin B₁₂ from R protein to intrinsic factor.

Clinical features of megaloblastic anaemia

- | | |
|-------------------------------|-----------------------|
| 1. Malaise | 2. Impotence |
| 3. Breathlessness | 4. Poor memory |
| 5. Paraesthesiae | 6. Depression |
| 7. Sore mouth | 8. Personality change |
| 9. Weight loss | 10. Hallucinations |
| 11. Altered skin pigmentation | |
| 12. Visual disturbance | |
| 13. Grey hair | |

Signs

- | | |
|----------------------|----------------------|
| 1. Smooth tongue | 2. Skin pigmentation |
| 3. Angular cheilosis | 4. Heart failure |
| 5. Vitiligo | 6. Pyrexia |

Investigations in megaloblastic anaemia

1. Haemoglobin—Often reduced, may be very low
2. MCV—usually raised, commonly > 120 fL
3. Erythrocyte count—Low for degree of anaemia
4. Blood film—Oval macrocytosis, poikilocytosis, red cell fragmentation, neutrophil hypersegmentation
5. Reticulocyte count—Low for degree of anaemia
6. Leucocyte count—Low or normal
7. Platelet count—Low or normal
8. Bone marrow—Increased cellularity, mega-

loblastic changes in erythroid series, giant metamyelocytes, dysplastic megakaryocytes, increased iron in stores, pathological non-ringing sideroblasts

9. Serum ferritin - Elevated
10. Plasma lactate - Elevated, often markedly

Folate deficiency

1. Folate are produced by plants and bacteria; hence dietary leafy vegetables (spinach, broccoli, lettuce), fruits (bananas, melons) and animal protein (liver, kidney) are a rich source.
2. An average Western diet contains more than the minimum daily intake of 50gm.
3. Excess cooking for longer than 15 minutes destroys folates.

Causes of folate deficiency

Diet—Poor intake of vegetables

Malabsorption—e.g. Coeliac disease

Increased demand—Cell proliferation, e.g. Hamolysis; Pregnancy

Drugs—Certain anticonvulsants (e.g. phenytoin); certain cytotoxic drugs (e.g. methotrexate)

Investigation to diagnose folic acid deficiency

1. Low serum folate levels (fasting blood sample)
2. Red cell folate levels low (but may be normal if folate deficiency is of very recent onset)
3. Macrocytic dysplastic blood picture
4. Megaloblastic marrow

Haemolytic anaemia

1. The normal red cell lifespan of 120 days may be shortened by a variety of abnormalities.
2. The bone marrow may increase its output of red cells six- to eight-fold by increasing the proportion of red cells produced, expanding the volume of active marrow and releasing reticulocytes prematurely.
3. If the rate of destruction exceeds this increased production rate, then anaemia will develop.
4. Red cell destruction overloads pathways for haemoglobin breakdown, causing a modest rise in unconjugated bilirubin in the blood and mild jaundice.

5. Increased reabsorption of urobilinogen from the gut results in an increase in urinary urobilinogen.
6. Activation of the bone marrow can result in a neutrophilia and immature granulocytes appearing in the blood to cause a leuco-erythroblastic blood film.
7. The appearance of the red cells may give an indication of the likely cause of the hemolysis.
8. Spherocytes are small, dark red cell which suggests autoimmune hemolysis or hereditary spherocytosis.
9. Sick cells suggest hemoglobinopathy
10. Red cell fragments indicate microangiopathic haemolysis
11. The compensatory erythroid hyperplasia may give rise to folate deficiency, when the blood findings will be complicated by the presence of megaloblastosis.
12. Measurement of red cell folate is unreliable in the presence of haemolysis and serum folate will be elevated.

Anemias

Classification of anemias according to morphology :

1. Hypochromic microcytic anemias
 - a. Iron deficiency anemia
 - b. Thalassemias, hemoglobinopathies
 - c. Sideroblastic anemia
 - d. Anemia of chronic disorders
 - e. Lead poisoning
 - f. Vitamin B₆ deficiency
2. Normochromic normocytic anemias
 - a. Anemia due to acute blood loss
 - b. Hemolytic anemias
 - c. Anemia due to bone marrow failure, renal disease, etc.
 - d. Anemia of chronic disorders
3. Macrocytic anemias
 - a. Megaloblastic anemia : Vitamin B₁₂ or folate deficiency.
 - b. Non-megaloblastic anemias : Liver disease, myelodysplasia, aplastic anemia, drug-induced anemia, hemolysis, hypothyroidism, alcoholism, etc.

Nutritional Anemias

Anemia is defined as a reduction below normal in the concentration of hemoglobin (Hb) or red blood cells in the blood. In India, hemoglobin levels below 13 g/dl in males, 12 g/dl in non-pregnant females, and 11 g/dl in pregnant females should be diagnosed as anemia.

Anemias may be classified based on red cell morphology as given below :

1. Microcytic hypochromic : Iron deficiency, thalassemias Chronic infection, sideroblastic anemia
2. Normocytic normochromic : Aplastic anaemia, post-haemorrhagic
3. Macrocytic normochromic : Folate deficiency, vitamin B₁₂ deficiency
4. Macrocytic hypochromic : Combined deficiency of iron and folate or B₁₂

The specific management depends on the etiological diagnosis described in the relevant chapters in this book and other text books of kayachikitsa vol I & II.

Thalassaemia

1. Thalassaemia is an inherited impairment of haemoglobin production, in which there is partial or complete failure to synthesise a specific type of globin chain.
2. In alpha-thalassaemia, disruption of one or both alleles on chromosome 16 may occur, with production of some or no alpha globin chains.
3. In beta-thalassaemia, defective production usually results from disabling point mutations causing no (3^0) or reduced (3^-) beta chain production.

Beta-thalassaemia

Failure to synthesise beta chains (beta-thalassaemia) is the most common type of thalassaemia, most prevalent in the Mediterranean area. Heterozygotes have thalassaemia minor, a condition in which there is usually mild anaemia and little or no clinical disability, which may be detected only when iron therapy for a mild microcytic anaemia fails. Homozygotes

(thalassaemia major) either are unable to synthesise haemoglobin A or at best produce very little; after the first 4-6 months of life they develop profound hypochromic anaemia.

Sickle-cell anaemia

1. It results from a single glutamic acid to valine substitution at position 6 of the beta globin polypeptide chain.
2. It is inherited as an autosomal recessive trait

Pathogenesis

When haemoglobin S is deoxygenated



The molecules of haemoglobin polymerise to form pseudocrystalline structures known as 'tactoids'.



These distort the red cell membrane



Produce characteristic sickle-shaped cells

Clinical features

1. Hypoxia
2. Acidosis
3. Dehydration
4. Infection.
5. Irreversibly sickled cells have a shortened survival and plug vessels in the microcirculation.
6. This results in a number of acute syndromes, termed 'crises', and chronic organ damage

Investigations

- Compensated anaemia, usually around 60-80g/h.
- The blood film shows sickle cells, target cells and features of hyposplenism.
- A reticulocytosis is present.
- The presence of HbS can be demonstrated by exposing red cells to a reducing agent such as sodium dithionite; HbA gives a clear solution, whereas HbS polymerises to produce a turbid solution.
- The definitive diagnosis requires haemoglobin electrophoresis to demonstrate the absence of

HbA, 2-20% HbF and the predominance of HbS.

- Both parents of the affected individual will have sickle-cell trait.

Leukaemias

1. Leukaemias are malignant disorders of the haematopoietic stem cell compartment, characteristically associated with increased numbers of white cells in the bone marrow and/or peripheral blood.
2. The course of leukaemia may vary from a few days or weeks to many years, depending on the type.

Epidemiology

The incidence of leukaemia of all types in the population is approximately 10/100 000 per annum, of which just under half are cases of acute leukaemia.

1. Males are affected more frequently than females
 - 3:2 in acute leukaemias
 - 2:1 in chronic lymphocytic leukaemia
 - 1.3:1 in chronic myeloid leukaemia.
2. Geographical variation in incidence does occur, the most striking being the rarity of chronic lymphocytic leukaemia in the Chinese and related races.
3. Acute leukaemia occurs at all ages.
4. Acute lymphoblastic leukaemia shows a peak of incidence in children aged 1-5 years.
5. All forms of acute myeloid leukaemia have their lowest incidence in young adult life and there is a striking rise over the age of 50.
6. Chronic leukaemias occur mainly in middle and old age.

Aetiology

1. The cause of the leukaemia is unknown in the majority of patients.
2. Risk factors for leukaemia

Ionising radiation

- After atomic bombing of Japanese cities (myeloid leukaemia)
- Radiotherapy for ankylosing spondylitis
- Diagnostic X-rays of the fetus in pregnancy

Cytotoxic drugs

- Especially alkylating agents (myeloid leukaemia, usually after a latent period of several years)
- Industrial exposure to benzene

Retroviruses

One rare form of T-cell leukaemia/lymphoma appears to be associated with a retrovirus similar to the viruses causing leukaemia in cats and cattle

Genetic

- Identical twin of patients with leukaemia
- Down's syndrome and certain other genetic disorders

Immunological

Immune deficiency states (e.g. hypogammaglobulinaemia)

Terminology and classification

Leukaemias are traditionally classified into four main groups:

- Acute lymphoblastic leukaemia (ALL)
- Acute myeloid leukaemia (AML)
- Chronic lymphocytic leukaemia (CLL)
- Chronic myeloid leukaemia (CML)

In acute leukaemia there is proliferation of primitive stem cells leading to an accumulation of blasts, predominantly in the bone marrow, which causes bone marrow failure.



In chronic leukaemia the malignant clone is able to differentiate, resulting in an accumulation of more mature cells.



Lymphocytic and lymphoblastic cells are those derived from the lymphoid stem cell (B cells and T cells).



Myeloid refers to the other lineages, i.e. precursors of red cells, granulocytes, monocytes and platelets.

Diagnosis of leukaemia

1. Blood test
Raised white count

2. Confirmed by examination of the bone marrow.

This includes the morphology of the abnormal cells, analysis of cell surface markers (immunophenotyping), clone-specific chromosome abnormalities and molecular changes.

3. The features in the bone marrow not only provide an accurate diagnosis but also give valuable prognostic information, allowing therapy to be tailored to the patient's disease.

Acute leukaemia

1. There is a failure of cell maturation in acute leukaemia.
2. Proliferation of cells which do not mature leads to an accumulation of useless cells which take up more and more marrow space at the expense of the normal haematopoietic elements.
3. Eventually this proliferation spills into the blood.
4. Acute myeloid leukaemia (AML) is about four times more common than acute lymphoblastic leukaemia (ALL) in adults.
5. In children the proportions are reversed, the lymphoblastic variety being more common.

The clinical features

Bone marrow failure features

1. Anaemia
2. Bleeding
3. Infection

Investigations

1. Blood examination
 - Anaemia with a normal or raised MCV
 - The leukocyte count may vary from as low as $1 \times 10^9/L$ to as high as $500 \times 10^9/L$ or more.
 - In the majority of patients the count is below $100 \times 10^9/L$.
 - Severe thrombocytopenia is usual but not invariable.
 - The appearance of blast cells in the blood film is usually diagnostic.
 - Sometimes the blast cell count may be very low in the peripheral blood and a bone marrow examination is necessary to confirm the diagnosis.

2. The bone marrow : Hypercellular, with replacement of normal elements by leukaemic blast cells in varying degrees of the cells. The presence of Auer rods in the cytoplasm of blast cells indicates a myeloblastic type of leukaemia.

Chronic myeloid leukaemia (CML)

1. It is a myeloproliferative stem cell disorder resulting in proliferation of all haematopoietic lineages but manifesting predominantly in the granulocytic series.
2. Maturation of cells proceeds fairly normally.
3. The disease occurs chiefly between the ages of 30 and 80 years, with a peak incidence at 55 years.
4. It is rare and accounts for 20% of all leukaemias.
5. The disease is found in all races.
6. Approximately 95% of patients with CML have a chromosome abnormality known as the Philadelphia (Ph) chromosome.

Natural history

The disease has three phases:

1. A chronic phase
Which the disease is responsive to treatment and is easily controlled, typically lasting 3-5 years
2. An accelerated phase (not always seen), in which disease control becomes more difficult.
3. *Blast crisis*, in which the disease transforms into an acute leukaemia, either myeloid (70%) or lymphoblastic (30%), which is relatively refractory to treatment. This is the cause of death in the majority of patients; therefore survival is dictated by the timing of blast crisis, which cannot be predicted.

Clinical features of chronic myeloid leukaemia

1. Tiredness
2. Weight loss
3. Breathlessness
4. Abdominal pain and discomfort
5. Lethargy
6. Anorexia
7. Sweating

8. Abdominal fullness
9. Bruising
10. Vague ill health

Physical examination

1. Splenomegaly is present in 90%; in about 10% the enlargement is massive, extending to over 15cm below the costal margin.
2. A friction rub may be heard in cases of splenic infarction.
3. Hepatomegaly occurs in about 50% Lymphadenopathy is unusual.

Investigations

1. Full blood count (FBC)
 - A normocytic, normochromic anaemia
 - The leucocyte count can vary from 10 to 600 $\times 10^9/L$.
 - In about one-third of patients there is a very high platelet count, sometimes as high as 2000 $\times 10^9/L$.
 - In the blood film the full range of granulocyte precursors from myeloblasts to mature neutrophils is seen but the predominant cells are neutrophils and myelocytes
 - Myeloblasts usually constitute less than 10% of all white cells.
 - There is often an absolute increase in eosinophils and basophils
 - Nucleated red cells are common.
2. Bone marrow
 - It should be obtained to confirm the diagnosis and phase of disease by morphology
 - Chromosome analysis to demonstrate the presence of the Ph chromosome
 - RNA analysis to demonstrate the presence of the BCR ABL gene product.
3. Other blood tests
 - Blood LDH levels are elevated
 - Uric acid level may be high due to increased cell breakdown.

Prolymphocytic leukaemia

1. This is a variant of chronic lymphatic leukaemia found mainly in males over the age of 60 years

2. 25 % of cases are of the T cell variety
3. There is massive splenomegaly with little lymphadenopathy
4. A very high leucocyte count, often in excess of $400 \times 10^9/L$
5. The characteristic cell is a large lymphocyte with a prominent nucleolus.
6. Treatment is generally unsuccessful and the prognosis very poor.
7. Leukapheresis, splenectomy and chemotherapy may be tried.

Hairy cell leukaemia

1. This is a rare chronic lymphoproliferative B cell disorder.
2. The male female ratio is 6:1
3. Median age at diagnosis is 50 years.
4. Presenting symptoms are those of general ill health and recurrent infection
5. Splenomegaly occurs in 90% but lymph node enlargement is unusual.
6. Severe neutropenia, monocytopenia and the characteristic hairy cells in the blood and bone marrow are typical.

Myelodysplastic syndrome (MDS)

1. This syndrome consists of a group of clonal haematopoietic disorders which represent steps in the progression to the development of leukaemia.
2. It presents with consequences of bone marrow failure (anaemia, recurrent infections or bleeding), usually in older people.
3. Median age at diagnosis is 69 years.
4. The blood film is characterised by cytopenias and abnormal-looking (dysplastic) blood cells, including macrocytic red cells and hypogranular neutrophils with nuclear hyper- or hyposegmentation.
5. The bone marrow is hypercellular with dysplastic changes in all three cell lines.
6. Blast cells may be increased but do not reach the 20% level which indicates acute leukaemia.
7. Chromosome analysis frequently reveals abnormalities, particularly of chromosome 5 or 7.

Thrombocytopenia

Clinical Significance

Determination of platelets is requested in the investigation of bleeding disorders.

Thrombocytopenia (decreased platelet count)- The count is less than 100,000 /cmm

The Mechanism leading to Thrombocytopenia

1. This may be due to decreased production of the bone marrow. This may be due :
 - A. Bone marrow failure.
 - B. Infiltration of the bone marrow by tumors or fibrosis.
2. Destruction or sequestration of the platelets by hypersplenism.
3. Antibodies destroying the platelets.
4. Destruction of the platelets by infection or drugs.
5. Increased utilization of disseminated intravascular coagulopathy.
6. In severe hemorrhage which leads to loss of platelets.
7. A large blood transfusion leads to a dilutional effect.

Causes of Thrombocytopenia

Physiological- New born, menstruating female

Pathological

- It is often associated with prolonged bleeding and poor clot retraction.
- It also occurs in aplastic anemia, pernicious megaloblastic hemolytic anemia
- Immune thrombocytopenia (ITP).
- Immune thrombocytopenic purpura (ITP)
- Hypersplenism.
- After a massive blood transfusion
- Infection like viral, and bacterial like HIV infection, dengue, dengue haemorrhaging fever, rubella infection
- Chemotherapy treatment.
- Leukemias, carcinoma, and myelofibrosis. This is due to the infiltration of the bone marrow.

- Disseminated intravascular coagulation (DIC).
- Toxemia of pregnancy, eclampsia.
- Antiplatelets antibody.
- Renal failure.
- Inherited diseases like Wiskott-Aldrich syndrome.
- An autoimmune disease like systemic lupus erythematosus

Thrombocytosis (increased platelet count)-

The count is more than 400,000 /cmm

Physiological- At high altitude, during severe exercise

Pathological

Secondary Thrombocytosis

1. Platelets are acute-phase reactants; therefore, platelet counts increase in response to various stimuli, including systemic infections, inflammatory conditions, bleeding, and tumors. This phenomenon is called reactive or secondary thrombocytosis, which is a benign form of thrombocytosis
2. Malignant tumors like leukemia, chronic myelogenous leukemia and lymphoma,
3. Polycythemia vera.
4. Iron deficiency anemia.
5. Autoimmune diseases like Rheumatoid arthritis, and Systemic lupus erythematosus (SLE).
6. Hodgkin's lymphoma
7. Chronic pancreatitis and inflammatory bowel disease.
8. Tuberculosis.
9. Infection and Inflammatory disorders
10. Postsplenectomy or hyposplenism
11. Trauma
12. Chronic inflammatory conditions
13. Hemorrhage, blood loss, or both
14. Rebound thrombocytosis
15. Asplenia (anatomic or functional)
16. Electric shock
17. Idiopathic

Causes of essential thrombocytosis (primary thrombocythemia)

- The etiology and predisposing factors for the development of essential thrombocytosis (primary thrombocythemia) remain unclear.
- Genetic transmission of this disorder is rare, although reports show several families with multiple members affected by essential thrombocytosis (primary thrombocythemia).
- Research suggests that a thrombopoietin production or receptor abnormality can cause familial essential thrombocytosis (primary thrombocythemia).

Thrombocythemia - When the count is above one million /cmm.

Causes

- Essential thrombocythemia is a type of chronic myeloproliferative disorder.
- It's not clear what causes this condition. About 90% of people with the disorder have a gene mutation contributing to the disease.

If an underlying condition such as an infection or iron deficiency causes a high platelet count, it's called secondary thrombocytosis. Compared with essential thrombocythemia, secondary thrombocytosis causes less risk of blood clots and bleeding.

MCQs

1. Which of the following is NOT a cause of microcytic anemia?
 - a. Thalassemia
 - b. Anemia of chronic disease
 - c. Iron deficiency anemia
 - d. Pancytopenia
2. Select the following that enhance Fe absorption (select all that apply)
 - a. Citric acid
 - b. Polyphenols (tea)
 - c. Phytate (bran)
 - d. Calcium
3. How would you diagnose aplastic anemia?
 - a. Blood smear
 - b. Bone marrow biopsy
 - c. Spleen biopsy
 - d. CBC

4. Most cases of acute leukaemia in children are of which type:

- a. T-cell ALL
- b. Mature B-cell ALL
- c. Hairy cell leukaemia
- d. Pre-B-cell ALL

5. Point mutation is characterized by which of the following disorders?

- a. Thalassemia
- b. Sickle cell anaemia
- c. Night blindness
- d. Down's syndrome

6. Beta-thalassemia is caused by which chromosomal mutation?

- a. 10
- b. 21
- c. 13
- d. 11

Answer

1. d	2. a	3. b
4. d	5. b	6. d